

# Xin Deng

Earth and Planets Laboratory, Carnegie Science  
5241 Broad Branch Road N.W., Washington DC 20015

E-mail: [xdeng@carnegiescience.edu](mailto:xdeng@carnegiescience.edu)

Website: [x1nd3ng.github.io](https://x1nd3ng.github.io)

## Appointments

2024–present **Carnegie Postdoctoral Fellow**, Earth and Planets Laboratory, Carnegie Science

## Education

2019–2024 **Ph.D.**, School of Earth and Space Sciences, University of Science and Technology of China

2015–2019 **B.Sc.**, School of the Gifted Young, University of Science and Technology of China

## Research interests

First-principles calculations & machine learning potentials

Properties of materials under high-pressure and high-temperature conditions

Equilibrium element partition and isotope fractionation

Structure and dynamics in the subduction zone

Composition and structure of the deep planetary interior

## Publications

### Articles: Submitted

[22] **Deng, Xin\***; Walter, Michael J.; Wu, Zhongqing; Cohen, Ronald E. "Cooling of the magma ocean forged Earth's mantle oxidation state", submitted to *Nature*.

[21] Zhao, Yajie; Hsu, Han; Wang, Wenzhong; Zhang, Xin; **Deng, Xin**; Chen, Bowen; Wu, Zhongqing\*. "Iron spin crossover in ferromagnesite (Mg,Fe)CO<sub>3</sub> and its link to lower-mantle seismic heterogeneity", submitted to *Geophysical Research Letters*.

[20] Liu, Cong\*; **Deng, Xin**; Cohen, Ronald E.\* "Order-Disorder in Fe-Si Alloys: Implications for Seismic Anisotropy and Thermal Evolution of Earth's Inner Core", submitted to *Science Advances*.

[19] **Deng, Xin**; Wu, Zhongqing\*; Song, Jian. "Water-Induced Mantle Overturms Leading to the Oxidation of Archean Upper Mantle", submitted to *Geophysical Research Letters*.

### Articles: Published or In-Press

[18] Park, Doyoon; **Deng, Xin\***; Deng, Jie\*. "Melting phase relation of seifertite and pyrite-type SiO<sub>2</sub> determined by machine learning potentials", *Physical Review B*, in press.

- [17] Yuan, Ye\*; Magali, John Keith; **Deng, Xin**; Rochira, Federica; Jiang, Xiuxuan; Thomas, Christine. "Revealing the nature of mid-mantle discontinuities through global interrogation of SS precursors versus P-to-S conversions", *Earth and Planetary Science Letters*, in press.
- [16] Liu, Hao\*; **Deng, Xin**; Leng, Wei\*; Wu, Zhongqing; Chu, Risheng; Wang, Xin. "Double volcanic tracks at Hawaii caused by bridgmanite-enriched primordial mantle blobs", *Nature Communications*, in press.
- [15] Zhang, Qiwei\*; Brenker, Frank E.; Timmerman, Suzette; Shirey, Steven B.; Stachel, Thomas; Luth, Robert W.; Locock, Andrew; Stern, Richard A.; Chinn, Ingrid L.; Nestola, Fabrizio; **Deng, Xin**; Pearson, D. Graham. (2026) "Sublithospheric diamond constraints on the state of deeply subducted slabs", *Science Advances*, 12(20), eadz7447. <https://doi.org/10.1126/sciadv.adz7447>
- [14] **Deng, Xin\***; Deng, Jie\*; Walter, Michael J.; Fei, Yingwei; Yang, Jing; Cohen, Ronald E. (2026) "Phase Relations and Melting of SiO<sub>2</sub> Under Mantle Conditions Determined from Machine Learning Potentials", *Geophysical Research Letters*, 53 (9). <https://doi.org/10.1029/2025GL119400>
- [13] Wang, Dong; Duan, Longyu; **Deng, Xin**; Wang, Wenzhong; Wu, Zhongqing\*. (2026) "Seismic Signature of the Upper Continental Crust: Implications from the thermoelastic properties of Liebermannite and K-hollandite II", *American Mineralogist*, 111 (2): 266-276. <https://doi.org/10.2138/am-2024-9562>
- [12] Zhao, Yajie; **Deng, Xin**; Wang, Wenzhong; Wu, Zhongqing\*; Yuan, Ye. (2025) "Seismic Visibility of carbonated subducted oceanic crust in the lower mantle", *Communications Earth & Environment*, 6, 146. <https://doi.org/10.1038/s43247-025-02128-0>
- [11] Yuan, Ye\*; Magali, John Keith; **Deng, Xin**; Sun, Daoyuan; Thomas, Christine. (2025) "Mantle discontinuities and reflectors beneath the Arctic Ocean and Aleutian-Alaska subduction zone: Evidence for MORB crust at the top of the lower mantle", *Earth and Planetary Science Letters*, 652, 119199. <https://doi.org/10.1016/j.epsl.2024.119199>
- [10] **Deng, Xin**; Chen, Yi-Xiang; Wang, Wenzhong; Li, Yonghui; Xiao, Zicong; Wu, Zhongqing\*. (2024) "Heavy magnesium isotopic signatures in arc lavas may be attributed to dehydration of subducting hydrated mantle", *Communications Earth & Environment*, 5, 299. <https://doi.org/10.1038/s43247-024-01466-9>
- [9] Zhao, Yajie; **Deng, Xin**; Chen, Ling; Wu, Zhongqing\*. (2024) "Is there a carbonated mid-lithosphere discontinuity in cratons?", *Journal of Geophysical Research: Solid Earth*, 129 (6). <https://doi.org/10.1029/2024JB028925>
- [8] Wang, Dong; Wu, Zhongqing\*; **Deng, Xin**. (2023) "Thermal conductivity of Fe-bearing bridgmanite and post-perovskite: Implications for the heat flux from the core", *Earth and Planetary Science Letters*, 621, 118368. <https://doi.org/10.1016/j.epsl.2023.118368>
- [7] **Deng, Xin**; Xu, Yinhan; Hao, Shangqin; Ruan, Youyi; Zhao, Yajie; Wang, Wenzhong; Ni, Sidao; Wu, Zhongqing\*. (2023) "Compositional and thermal state of the lower mantle from joint 3D inversion with seismic tomography and mineral elasticity", *Proceedings of the National Academy of Sciences*, 120 (26). <https://doi.org/10.1073/pnas.2220178120>
- [6] **Deng, Xin**; Song, Jian; Qian, Wangsheng; Wu, Zhongqing\*. (2023) "Seismic signals induced by the Metasomatism of mantle wedge by siliceous melts: Insights from the elasticity of orthopyroxene at high

pressure and temperature", *Tectonophysics*, 846, 229681. <https://doi.org/10.1016/j.tecto.2022.229681>

[5] Wang, Dong; Wu, Zhongqing\*; **Deng, Xin**. (2022) "Thermal Conductivity of Hydrous Wadsleyite Determined by Non-Equilibrium Molecular Dynamics Based on Machine Learning", *Geophysical Research Letters*, 49 (22). <https://doi.org/10.1029/2022GL100337>

[4] **Deng, Xin**; Luo, Chenxing; Wentzcovitch, Renata M.; Abers, Geoffrey A.; Wu, Zhongqing\*. (2022) "Elastic Anisotropy of Lizardite at Subduction Zone Conditions", *Geophysical Research Letters*, 49 (18). <https://doi.org/10.1029/2022GL099712>

[3] Zhao, Yajie; Wu, Zhongqing\*; Hao, Shangqin; Wang, Wenzhong; **Deng, Xin**; Song, Jian. (2022) "Elastic properties of Fe-bearing Akimotoite at mantle conditions: Implications for composition and temperature in lower mantle transition zone", *Fundamental Research*, 2, 570 577. <https://doi.org/10.1016/j.fmre.2021.12.013>

[2] Liu, Cong; Wang, Junjie; **Deng, Xin**; Wang, Xiaomeng; Pickard, Chris J.; Helled, Ravit; Wu, Zhongqing; Wang, Huitian; Xing, Dingyu; Sun, Jian\*. (2022) "Partially Diffusive Helium-Silica Compound under High Pressure", *Chinese Physics Letters*, 39, 076101. <https://doi.org/10.1088/0256-307X/39/7/076101>

[1] Luo, Chenxing; **Deng, Xin**; Wang, Wenzhong; Shukla, Gaurav; Wu, Zhongqing; Wentzcovitch, Renata M.\* (2021) "cij: A Python code for quasiharmonic thermoelasticity", *Computer Physics Communications*, 267, 108067. <https://doi.org/10.1016/j.cpc.2021.108067>

## Invited talks

- 2026.5    **Melting in Planetary Interiors: Constraints from Large-Scale Atomistic Simulations**  
EPL, Carnegie Science (Invited by Ronald E. Cohen)
- 2026.3    **Melts in Early Earth: Large-Scale Modeling of Planetary Materials**  
Department of Geosciences, Stony Brook University (Invited by Deanne Rogers)
- 2025.9    **Magma Ocean Evolution Shapes the Redox State of Planets**  
National Museum of Natural History, Smithsonian Institution (Invited by Elizabeth Cottrell)
- 2025.1    **Subduction Zone Structure and Lower Mantle Composition based on First-Principles Calculations**  
Guangzhou Institute of Geochemistry (Invited by Yangfan Deng)
- 2023.12    **Large Low Shear Velocity Provinces**  
USTC (Recent Advances in Geophysics)
- 2023.11    **Physics and Chemistry of the Earth's Interior: Insights from ab-initio Calculations**  
USTC (Academic Salon for Postgraduates of Solid Geophysics)
- 2022.4    **Formation of Low  $V_P/V_S$  Regions in the Mantle Wedges of Subduction Zones**  
USTC (Academic Forum for Postgraduates of Solid Geophysics)
- 2021.12    **3D Compositional and Thermal State of the Lower Mantle**  
USTC (Academic Forum for Postgraduates of Solid Geophysics)

## Conference presentations

- [34] **Deng, Xin**; Walter, Michael; Wu, Zhongqing; Cohen, Ronald E. (2026) "Quantifying Iron Disproportionation in Silicate Melts Using Machine Learning Potentials", **24th IMA General Meeting**, Nanjing, China.
- [33] **Deng, Xin**; Walter, Michael; Wu, Zhongqing; Cohen, Ronald E. (2026) "Magma ocean cooling controls the mantle oxidation state", **Goldschmidt 2026 Conference**, Montreal, Canada.
- [32] **Deng, Xin**; Walter, Michael; Wu, Zhongqing; Cohen, Ronald E. (2025) "Secular cooling of the Magma Ocean Controls Early Planetary Oxidation State", **AGU Annual Meeting 2025**, New Orleans, USA. (poster)
- [31] **Deng, Xin**; Walter, Michael; Wu, Zhongqing; Cohen, Ronald E. (2025) "Secular cooling of the Magma Ocean Controls Early Planetary Oxidation State", **The 10<sup>th</sup> International Symposium on "From Atom to Earth" High pressure Science**, Beijing, China.
- [30] **Deng, Xin**; Zhao, Yajie; Walter, Michael; Wu, Zhongqing. (2025) "Elasticity of Aragonite under mantle conditions: implications for the carbonation of the big mantle wedge", **Goldschmidt 2025 Conference**, Prague, Czech.
- [29] **Deng, Xin**; Zhao, Yajie; Wu, Zhongqing. (2024) "Elasticity of Aragonite under mantle conditions by first-principles calculations", **AGU Annual Meeting 2024**, Washington, USA.
- [28] **Deng, Xin**; Chen, Yi-Xiang; Wang, Wenzhong; Li, Yonghui; Xiao, Zicong; Wu, Zhongqing. (2023) "Dehydration of Subducting Hydrated Mantle Revealed by Heavy Mg Isotopes in Arc Lavas", **AGU Fall Meeting 2023**, San Francisco, USA. (poster)
- [27] **Deng, Xin**; Chen, Yi-Xiang; Wang, Wenzhong; Li, Yonghui; Xiao, Zicong; Wu, Zhongqing. (2023) "Dehydration of Subducting Oceanic Mantle indicated by Heavy Magnesium Isotopes in Arc Lavas", **Annual Meeting of Chinese Geoscience Union 2023**, Zhuhai, China.
- [26] **Deng, Xin**; Xu, Yinhan; Hao, Shangqin; Ruan, Youyi; Zhao, Yajie; Wang, Wenzhong; Ni, Sidao; Wu, Zhongqing. (2023) "3D Composition and Temperature Structure of the Lower Mantle: Based on Seismic Tomography and Mineral Physics", **Seminar on Deep Mantle and Core**, Beijing, China.
- [25] **Deng, Xin**; Chen, Yi-Xiang; Wang, Wenzhong; Li, Yonghui; Xiao, Zicong; Wu, Zhongqing. (2023) "Dehydration of Subducting Hydrated Mantle indicated by High  $\delta^{26}\text{Mg}$  Values in Arc Lavas", **Habitable Earth – Geoscience for Sustainability**, Qingdao, China.
- [24] **Deng, Xin**; Xu, Yinhan; Hao, Shangqin; Ruan, Youyi; Zhao, Yajie; Wang, Wenzhong; Ni, Sidao; Wu, Zhongqing. (2023) "Combining the high P-T elasticity of lower mantle minerals with seismic tomography to Constrain the 3D Composition and Temperature Structure of the Lower Mantle", **The 9<sup>th</sup> "From Atom to Earth" Symposium on High-pressure Science**, Chengdu, China. (**Best Student Presentation Award (Hongsen Award)**)
- [23] **Deng, Xin**; Xu, Yinhan; Hao, Shangqin; Ruan, Youyi; Zhao, Yajie; Wang, Wenzhong; Ni, Sidao; Wu, Zhongqing. (2023) "Construction of 3D Composition and Temperature Structure of the Lower Mantle", **The 2nd conference of "China Seismological Reference Model"**, Hefei, China. (poster)
- [22] **Deng, Xin**; Xu, Yinhan; Hao, Shangqin; Ruan, Youyi; Zhao, Yajie; Wang, Wenzhong; Ni, Sidao; Wu, Zhongqing. (2023) "Compositional and Thermal State of the Lower Mantle from Joint 3D Inver-

- sion with Seismic tomography and Mineral Elasticity", *Goldschmidt 2023 Conference*, Lyon, France.
- [21] **Deng, Xin**; Xu, Yinhan; Hao, Shangqin; Ruan, Youyi; Zhao, Yajie; Wang, Wenzhong; Ni, Sidao; Wu, Zhongqing. (2023) "Compositional and Thermal State of the Lower Mantle from Joint 3D Inversion with Seismic tomography and Mineral Elasticity", *2023 IPACES Annual Meeting*, Hefei, China. (poster / **Poster Presentation Award 2<sup>nd</sup> Prize**)
- [20] **Deng, Xin**; Xu, Yinhan; Hao, Shangqin; Ruan, Youyi; Zhao, Yajie; Wang, Wenzhong; Ni, Sidao; Wu, Zhongqing. (2023) "Compositional and Thermal State of the Lower Mantle from Joint 3D Inversion with Seismic Tomography and Mineral Elasticity", *Japan Geoscience Union Meeting 2023*, Chiba, Japan. (poster / **Outstanding Student Presentation Award**)
- [19] **Deng, Xin**; Xu, Yinhan; Hao, Shangqin; Ruan, Youyi; Zhao, Yajie; Wang, Wenzhong; Ni, Sidao; Wu, Zhongqing. (2023) "Inversion of 3D Thermo-Chemical Structure of the Lower Mantle using Seismic Tomography and Elasticity of Lower-mantle Minerals at High Pressure and Temperature Conditions", *The 21<sup>st</sup> China High-Pressure Science Conference*, Dalian, China.
- [18] **Deng, Xin**; Xu, Yinhan; Hao, Shangqin; Ruan, Youyi; Zhao, Yajie; Wang, Wenzhong; Ni, Sidao; Wu, Zhongqing. (2023) "3D Compositional and Thermal Structure of the Lower Mantle Inverted from Seismic Tomography and Mineral Elasticity", *The 8<sup>th</sup> Young Scientist Forum of Earth Science*, Wuhan, China.
- [17] **Deng, Xin**; Xu, Yinhan; Hao, Shangqin; Ruan, Youyi; Zhao, Yajie; Wang, Wenzhong; Ni, Sidao; Wu, Zhongqing. (2023) "3D Thermo-Chemical Structure of the Lower Mantle based on Seismic Tomography and Mineral Elasticity", *The 5<sup>th</sup> Congress of China Geodesy and Geophysics*, Wuhan, China.
- [16] **Deng, Xin**; Luo, Chenxing; Wentzcovitch, Renata; Abers, Geoffrey; Wu, Zhongqing. (2023) "Elasticity of Serpentine Minerals at Subduction Zone Conditions", *The 18<sup>th</sup> Annual Meeting of Chinese Society for Mineralogy Petrology and Geochemistry*, Hefei, China.
- [15] **Deng, Xin**; Song, Jian; Qian, Wangsheng; Wu, Zhongqing. (2023) "Elasticity of Orthopyroxene at High P-T Conditions: Implications for the Metasomatism of Mantle Wedge by Siliceous Melts", *The 18<sup>th</sup> Annual Meeting of Chinese Society for Mineralogy Petrology and Geochemistry*, Hefei, China.
- [14] **Deng, Xin**; Xu, Yinhan; Hao, Shangqin; Ruan, Youyi; Zhao, Yajie; Wang, Wenzhong; Ni, Sidao; Wu, Zhongqing. (2022) "Compositional and Thermal State of the Lower Mantle from Joint 3D Inversion with Seismic Tomography and Mineral Elasticity", *AGU Fall Meeting 2022*, Chicago, USA. (**Outstanding Student Presentation Award**)
- [13] **Deng, Xin**; Xu, Yinhan; Hao, Shangqin; Ruan, Youyi; Zhao, Yajie; Wang, Wenzhong; Ni, Sidao; Wu, Zhongqing. (2022) "Compositional and Thermal State of the Lower Mantle from Joint 3D Inversion with Seismic Tomography and Mineral Elasticity", *Annual Meeting of Chinese Geoscience Union 2022/2021*, Fuzhou, China.
- [12] **Deng, Xin**; Song, Jian; Qian, Wangsheng; Wu, Zhongqing. (2022) "Seismic signals induced by the Metasomatism of mantle wedge by siliceous melts: insights from the Elasticity of Orthopyroxene at High Pressure and Temperature", *Annual Meeting of Chinese Geoscience Union 2022/2021*, Fuzhou, China. (**Outstanding Student Presentation Award**)
- [11] **Deng, Xin**; Xu, Yinhan; Hao, Shangqin; Ruan, Youyi; Zhao, Yajie; Wang, Wenzhong; Ni, Sidao;

- Wu, Zhongqing. (2022) "Compositional and Thermal State of the Lower Mantle from Joint 3D Inversion with Seismic tomography and Mineral Elasticity", ***Frontiers of High Pressure Research – Science under Extreme Conditions***, Shanghai, China. (poster / **Best Poster Award**)
- [10] **Deng, Xin**; Luo, Chenxing; Wentzcovitch, Renata; Abers, Geoffrey; Wu, Zhongqing. (2022) "Elastic anisotropy of lizardite at subduction zone conditions", ***The 33<sup>rd</sup> IUPAP Conference on Computational Physics***, Austin, USA.
- [9] **Deng, Xin**; Song, Jian; Qian, Wangsheng; Wu, Zhongqing. (2022) "Elasticity of Orthopyroxene at High Pressure and Temperature: Insights into the Metasomatism of Mantle Wedge by Siliceous Melts", ***Goldschmidt 2022 Conference***, Honolulu, USA. (poster)
- [8] **Deng, Xin**; Song, Jian; Qian, Wangsheng; Wu, Zhongqing. (2022) "Elasticity of Orthopyroxene at High Pressure and Temperature: Insights into the Metasomatism of Mantle Wedge by Siliceous Melts", ***Japan Geoscience Union Meeting 2022***, Chiba, Japan.
- [7] **Deng, Xin**; Song, Jian; Qian, Wangsheng; Wu, Zhongqing. (2021) "Elasticity of Orthopyroxene at High Pressure and Temperature: Insights into the Metasomatism of Mantle Wedge by Siliceous Melts", ***AGU Fall Meeting 2021***, New Orleans, USA. (poster)
- [6] **Deng, Xin**; Luo, Chenxing; Wentzcovitch, Renata; Abers, Geoffrey; Wu, Zhongqing. (2021) "Elasticity of Lizardite at High Pressure and Temperature: Implications for the Water Content in Subduction Zones", ***AGU Fall Meeting 2021***, New Orleans, USA. (poster)
- [5] **Deng, Xin**; Song, Jian; Qian, Wangsheng; Wu, Zhongqing. (2021) "Elasticity of Orthopyroxene at High Pressure and Temperature: Insights into the Metasomatism of Mantle Wedge by Siliceous Melts", ***2021 International Symposium on Deep Earth Exploration and Practices***, Nanjing, China.
- [4] **Deng, Xin**; Luo, Chenxing; Wentzcovitch, Renata; Wu, Zhongqing. (2021) "Elasticity of Lizardite at High Pressure and Temperature: Implications for the Water Content in Subduction Zones and Arc Magmas", ***The 6<sup>th</sup> Conference on Earth System Science***, Shanghai, China.
- [3] **Deng, Xin**; Song, Jian; Qian, Wangsheng; Wu, Zhongqing. (2021) "Elasticity of Orthopyroxene at High Pressure and Temperature: Insights into the Metasomatism of Mantle Wedge by Siliceous Melts", ***The 8<sup>th</sup> "From Atom to Earth" Symposium on High-pressure Science and Earth Science***, Guiyang, China.
- [2] **Deng, Xin**; Wu, Zhongqing. (2021) "Compositional and thermal state of the Moon from first-principles calculations", ***2021 National Planetary Science Conference***, Suzhou, China.
- [1] **Deng, Xin**; Wu, Zhongqing. (2020) "Elasticity of Lizardite at High Pressure and Temperature: Implications for the Water Content in Subduction Zones", ***Annual Meeting of Chinese Geoscience Union 2020***, Chongqing, China.

## Honors

|           |                                                                                                                          |
|-----------|--------------------------------------------------------------------------------------------------------------------------|
| 2024      | <b>Carnegie Postdoctoral Fellowship</b> (Earth and Planets Lab, Carnegie Science)                                        |
| 2024      | <b>Excellent Graduation Thesis Award</b> (USTC)                                                                          |
| 2024      | <b>Excellent Graduate of Anhui Province Ordinary Colleges and Universities</b> (Anhui Province)                          |
| 2024      | <b>Outstanding Graduate of University of Science and Technology of China</b> (USTC)                                      |
| 2024      | <b>President's Award of the Chinese Academy of Sciences</b> (Chinese Academy of Sciences)                                |
| 2024      | <b>Dean's Award for Outstanding Students, School of Earth and Space Sciences</b> (USTC)                                  |
| 2023      | <b>National Scholarship for Postgraduates</b> (USTC)                                                                     |
| 2023      | <b>Outstanding Student Presentation Award</b> (Seminar on Deep Mantle and Core)                                          |
| 2023      | <b>Best Student Presentation Award (Hongsen Award)</b> (The 9th "From Atom to Earth" Symposium on High-pressure Science) |
| 2023      | <b>Poster Presentation Award 2<sup>nd</sup> Prize</b> (2023 IPACES Annual Meeting)                                       |
| 2023      | <b>Outstanding Student Presentation Award</b> (Japan Geoscience Union Meeting 2023)                                      |
| 2022      | <b>Outstanding Student Presentation Award</b> (AGU Fall Meeting 2022)                                                    |
| 2022      | <b>Outstanding Student Presentation Award</b> (Annual Meeting of CGU 2022/2021)                                          |
| 2022      | <b>Best Poster Award</b> (Frontiers of High Pressure Research – Science under Extreme Conditions)                        |
| 2021-2023 | <b>First-Class Academic Scholarship for Doctoral Students</b> (USTC)                                                     |
| 2019-2020 | <b>First-Class Academic Scholarship for Master Students</b> (USTC)                                                       |

## Teaching experience

### Teaching Assistant at University of Science and Technology of China

|           |           |                    |
|-----------|-----------|--------------------|
| 2021–2023 | GEPH6411P | Solid Mechanics    |
| 2018–2023 | PHYS1001A | Mechanics A        |
| 2019–2020 | PHYS1004A | Electromagnetism A |

## Service

**Peer review** for Communications Earth & Environment (1), Physics of the Earth and Planetary Interiors (1)

**Proposal reviewer** (non-panelist) for NSF (1)